Understanding Robotaxi Operations Through Systems Psychology: A Case Study of San Francisco

Asha Golveo

Embry-Riddle Aeronautical University

MSHF 641 Systems Psychology

Dr. Clint R. Balog

March 8, 2026

Abstract

Autonomous vehicle (AV) technology is rapidly transforming urban transportation, with robotaxis representing one of the first large-scale commercial deployments of Level 4 automation. Unlike privately owned AVs, robotaxis operate as cloud-connected ridesharing fleets within defined operational design domains, integrating advanced sensing technologies, artificial intelligence, remote human oversight, and regulatory governance. This paper analyzes the deployment of robotaxi services in San Francisco as a complex sociotechnical system, where technological capability must align with human cognition, organizational structures, regulatory institutions, and urban infrastructure. Using a systems psychology perspective, the study examines how interactions between human, technical, organizational, and environmental subsystems shape system performance, safety, and public acceptance. The analysis highlights key inflection points in the evolution of the San Francisco robotaxi ecosystem, including the 2023 suspension of Cruise's operating permit following a high profile pedestrian accident, Waymo's subsequent expansion, growing community resistance, and the December 2025 citywide power outage that caused hundreds of Waymos to stall simultaneously. These events illustrate systemic vulnerabilities related to automation boundaries, remote operator scalability, regulatory fragmentation, and infrastructure dependencies. Through stakeholder mapping, behavioral system dynamics, and functional decomposition, the paper identifies reinforcing and balancing feedback loops influencing system stability, including technology learning effects, trust erosion following incidents, fleet scaling risks, and regulatory intervention mechanisms.

Understanding Robotaxi Operations Through Systems Psychology: A Case Study of San Francisco

An autonomous vehicle (AV), also known as a self-driving car, is capable of operating with minimal or no human input. They are equipped with sensors, cameras, radar, and AI software that allows them to perceive their environment and control driving without human intervention. The Society of Automotive Engineers (SAE) assigns automation levels from Level 0 to Level 5, with 0 representing no automation, and 5 representing fully autonomous with no human driver needed (Synopsys Editorial Staff, 2025). Tesla's self-driving system is classified as Level 2 partial automation, with the system having the ability to automate navigation, steering, and speed, but require human supervision to manage steering, braking, and monitoring the environment. Level 3 represents conditional automation, with the system responsible for short periods of driving but may request human intervention, and Level 4 represents high automation with AVs operating in self-driving mode within designated areas.

Robotaxis are Level 4 AVs that operate as a ridesharing service, providing on-demand transportation by allowing passengers to request a ride using a mobile app. The robotaxis operate within predefined urban boundaries with extensive HD mapping, while human operators monitor fleets, and provide guidance in unfamiliar situations. The fleet of cars are managed by a cloud-connected system that coordinates dispatch, ride-hailing integration, and diagnostics. Robotaxi systems are not purely technical systems, they are complex adaptive sociotechnical ecosystems requiring alignment between automation capability, human cognition, organizational policy, and public infrastructure (Fraedrich et al., 2015). Robotaxi adoption is vulnerable to technical perception failures, human-automation boundary design, trust calibration, management of rare events, and sociotechnical integration. Waymo, a subsidiary of Alphabet/Google, is the most significant provider in the US, operating in the San Francisco Bay Area, Phoenix, Los Angeles, Atlanta, Austin, and Miami (Wikipedia, 2025). San Francisco has served as a significant testing site for robotaxi deployment, with companies such as Waymo, Cruise, and Zoox all receiving authorization from the California Public Utilities Commission (CPUC) to operate its autonomous vehicle passenger service. The city of San Francisco presents high system complexity due to high pedestrian and cyclist traffic, steep hills, frequent construction, complex right-of-way interactions, and emergency vehicle navigation.

Key Incidents and Inflection Points

1. October 2023 - Cruise Permit Suspension following a severe incident in which a Cruise robotaxi struck and dragged a pedestrian who had been hit by another vehicle. The California DMV suspended Cruise's operating permits. This marked the first major regulatory reversal in the AV industry.
2. 2022–2024 - Waymo’s continued expansion despite Cruise's setback, expanding its commercial paid service across San Francisco.
3. Ongoing - Community Resistance: San Francisco firefighters, transit unions, disability rights advocates, and neighborhood groups mounted organized opposition, citing emergency vehicle interference, labor displacement, and equity concerns.
4. 2023–2025 — Regulatory Evolution: CPUC granted expanded commercial operating authority while simultaneously developing new incident reporting and data-sharing requirements, reflecting an adaptive but reactive regulatory posture
5. December 2025 - A power outage in San Francisco led to hundreds of robotaxis stalling mid-intersection, blocking emergency responders, and overwhelming the remote operator

system. This highlighted the tension of a technology designed to replace human judgement, yet still depends on human oversight infrastructure that does not scale at the same rate as the fleet.

System Components

Human Subsystem

Human actors are integral to the robotaxi sociotechnical system, with passenger adoption determining viability, remote operators managing safety, fleet operations staff maintaining vehicles, and the implicit control of robotaxi behavior by other road users (cyclists and other drivers). Passenger experience is initially shaped by their interaction with the mobile app, then trust dictates perception of comfort and safety when the presence of a human driver is not available to act as a social safety agent. Mental models shape user expectations and trust formation, informing behaviours, decisions, and system interaction (Svenson & Eriksson, 2017).

Remote operators are the critical human-in-the-loop component, providing guidance regarding route suggestions, and contextual instructions when vehicles encounter rare and complicated situations (Schrank et al., 2025). Remote operators do not take direct steering control, but passengers can interact with remote operators by requesting information about vehicle operations. Remote operators are limited by operational challenges, including incomplete data from sensors, interface design of control centers, and latency when receiving information from the AVs. Furthermore, remote operators must manage mental resources when alternating between AVs, and impaired situational awareness when the AV has direct responsibility for driving tasks (Meir et al., 2025).

Fleet operations staff must conduct preventative and corrective maintenance for vehicle and sensor hardware, including management of charging infrastructure and daily recharging of the electric vehicles. Pedestrians, cyclists, and human drivers are indirect human agents that impact traffic flow and robotaxi behavior, utilizing mental models to interpret vehicle intent. The technical-human interface between the onboard AI and remote fleet operators is a critical organizational boundary, where the responsibility shifts from machine perception to human judgment.

Technical Subsystem

Most robotaxis utilize LiDAR (Light Detection and Ranging), a remote sensing method to create precise 3D maps and models of the surrounding environment. High-definition cameras provide color and texture, allowing the system to read traffic lights, signs, and distinguish between objects of the same color. Radar is used to detect the velocity and distance of objects, excelling in environments with fog, rain, or snow, when cameras and LiDAR may degrade in functionality. The software stack of robotaxis includes AI that mimics human cognition of a human driver by managing perception and localization. The subcomponents include the prediction system that analyzes probable paths of every moving object, the planning system that selects vehicle speed, lane position, and maneuvering, and the actuation system that executes steering, braking, or engaging motors (Grigorescu et al., 2020). Key inputs include real-time sensor streams, HD map tiles, traffic signal state, remote operator commands, and software updates. Key outputs include vehicle motion, passenger transport outcomes, incident telemetry, and training data contributions to the fleet learning systems.

Organizational Subsystem

The organizational subsystem maintains the "Operational Design Domain" (ODD), which defines the geofenced areas where vehicles can operate. Robotaxis operate within this rigid physical boundary restricted by the geofence, preventing vehicles from functioning outside of

HD mapped zones. This boundary is a point of tension between tech companies and city planners regarding "transportation equity", where the Geofence creates service desserts that make areas invisible to the system (Jomehpour Chahar Aman & Smith-Colin, 2020). Furthermore, robotaxi companies must navigate regulatory and legal boundaries that mandate vehicle safety standards, state commercial operations, and local enforcement by city governments. The regulatory component comprises three distinct institutions with overlapping mandates: the California DMV (vehicle safety certification, permit issuance and revocation, incident investigation); the California Public Utilities Commission (commercial transportation authority, consumer protection, service equity requirements); and the City and County of San Francisco (street use permits, emergency services coordination, zoning, and political representation of affected communities). While the City of San Francisco has the inability to directly regulate AV operations, regulators also operate at a disadvantage when they must rely on data that companies choose to report.

Environmental Subsystem

In a sociotechnical system, the environmental boundary can be defined as the point when the robotaxi system interacts with external, uncontrollable physical forces. This component encompasses the urban infrastructure of San Francisco that the robotaxi system operates within and depends upon, including the street network of road markings, traffic signals, cellular networks, and electrical grids; the San Francisco Fire Department and SFPD emergency response systems, the San Francisco Municipal Transportation Agency (Muni, BART interfaces, traffic management); and the physical environments of construction zones, crosswalks, bus stops, loading zones. Infrastructure changes (construction, signal retiming, new crosswalks) occur continuously, but are not systematically communicated to robotaxi operators in real time, leading to mismatches between the physical environment and the robotaxi's internal map representation.

Weather is a volatile environmental boundary that can reduce sensor accuracy, with extreme conditions causing robotaxis to pull over or stop service due to safety concerns. Weather conditions can also impact road conditions, with wet and icy roads increasing the danger of vehicle hydroplaning. Extreme wind can lead to fallen trees and debris on the road, obstructing the vehicle from moving, and causing delayed traffic during AV recalibration. However, the most consequential boundary stress in this component is the interaction between robotaxi operations and emergency response. When a robotaxi stops in a travel lane during an emergency, conflict arises between the boundaries of the AV system and emergency services. While emergency services lack authority over AV behaviour, AV systems lack reliable real-time awareness of emergency vehicle approach paths.

Stakeholder Mapping

Stakeholder	Primary Interests	System Position	Key Tensions
Waymo / Google (Alphabet)	Commercial scale, regulatory approval, safety record, brand trust	Core operator; most advanced technology; relatively cautious deployer	Speed vs. safety; corporate secrecy vs. public accountability
Cruise / GM	Market share, commercial revenue, investor returns	Core operator; aggressive growth strategy; suffered major reputational crisis	Competitiveness vs. safety culture; data transparency
Passengers	Safe, affordable, convenient transport	Trust in technology vs. anxiety about lack of human driver	Benefits (safety, mobility) vs. disruptions (blocking, privacy, aesthetics)
California DMV	Public safety, regulatory compliance, AV governance	Primary vehicle safety regulator; permit-granting authority	Speed of permit issuance vs. verification rigor
California CPUC	Safe, compliant commercial deployment; consumer protection	Commercial transportation authority; shared jurisdiction with DMV	Overlapping mandates; regulatory uncertainty
City of San Francisco	Traffic management, emergency services, public space, equity	Most affected jurisdiction; limited formal AV authority	Lack of local control; infrastructure impact; community pressure
San Francisco Fire Dept.	Emergency response effectiveness, public safety	Critical infrastructure partner; incident responder	AV obstruction of emergency vehicles; data access limitations

Feedback Loops and System Dynamics

Reinforcing Loops

R1. Technology Learning Loop

As robotaxis drive more miles, they generate more algorithm training data and increased rare scenario encounters, improving system performance and passenger trust. Rapid AV expansion accelerates technology maturity, thereby increasing regulatory confidence and expanded operating permissions. However, incidents during the learning phase can trigger a balancing loop that reverses the cycle entirely.

R2. Trust Erosion After Incidents

High profile failures and incidents produce negative public perception, creating political pressure on regulators and permit restrictions. This can lead to reduced passenger trust and a reduction in rides, leading to a reduction in learning and higher probability of failures. The Cruise incident in 2023 demonstrates how damaged trust can activate this loop, eventually destroying Cruise’s business.

R3. Fleet Scale and Systemic Risk

Large fleets of robotaxis create greater fleet density in San Francisco. When the system is experiencing a system failure, the incident affects many vehicles and becomes a city-wide event. This can exacerbate traffic and emergency conditions, leading to a larger political and public backlash. The growth that feeds the technology learning loop can simultaneously increase risk.

Balancing Loops

B1. Regulatory Constraint

Fleet incidents prompt regulatory intervention, which reduces operating permissions and reduces incident exposure. This becomes a motivator for reauthorization and drives technological improvements, thereby improving safety and enabling reauthorization. This balancing loop is the primary corrective mechanism. Its effectiveness depends on regulatory capacity, information quality, and the time delays between incidents and substantive interventions.

B2. Public Acceptance and Trust

AV deployment generates observable benefits of accessible transport and reduced DUI-related accidents. This benefit can increase public awareness, reduce novelty anxiety and produce greater acceptance, which generates political support for expansion.

Emergent Behaviors

Technical-Social Interactions

AVs were programmed with a safety measure to exercise extreme caution and stop for any ambiguous obstacle. Human drivers and pedestrians have learned to exploit this feature, with pedestrians intentionally stepping off curbs to halt vehicles, cyclists cutting in front of robotaxis, and human drivers merging aggressively around AVs, which they know must yield. Şahin et al. (2022) found that pedestrians do exhibit deviant behavior of jaywalking and AV bullying when encountering conflict avoidant AVs. These patterns can create a downstream emergence, shifting traffic flow and other drivers' expectations.

Public robotaxi resistance generated a cascade of emergent behaviors, including coning and subsequent modification in detection algorithms. Coning behavior, placing a cone on an AV hood to exploit its object detection and immobilize it, arose from individuals discovering the technical vulnerability and using it as a tool for resistance. Circulation on social media led to an increase in incidents, with AV companies modifying detection algorithms, which in turn changed vehicle behavior and legitimate road interactions. A local act of street level protest propagated into corporate engineering decisions, regulatory attention, and public discourse about AV legitimacy.

Economic Emergence

A significant emergent property is the effect on city traffic and infrastructure. In a dense city like San Francisco, it is often cheaper to keep a robotaxi moving rather than pay for parking. This leads to constant cruising by robotaxis, which increases road wear and traffic, even when vehicles are not transporting a passenger. Furthermore, the ability to sleep and work in a robotaxi can encourage an emergent shift in residential patterns, allowing residents more options to live outside of city centers.

Functional Decomposition

F1 Autonomous Mobility: Environment perception, prediction and planning, vehicle control, passenger interface. The technical core of the system.

F2 Fleet Operations: Remote monitoring, dispatch and routing, vehicle maintenance, data management. The supervisory layer connecting the technical stack to the organisational envelope.

F3 Regulatory Compliance: Permitting, incident reporting, liability management. The functions maintain legal operating authority.

F4 Social and Community Interface: Public trust management, street-level interaction (including adversarial behaviour response and emergency services coordination), labour relations. These functions were the least developed and most consequential for system viability.

F5 Learning and Improvement: Model development, operational learning from incidents, infrastructure adaptation. The system's capacity for self-correction over time.

The functional decomposition highlights that F4 (Social and Community Interface) sits at the same hierarchical level as F1 (Autonomous Mobility).

Systemic Recommendations

Technical System Resilience

1. Mandate and verify real-time integration with emergency dispatch systems to pre-route vehicles away from active incident zones. Mandating developed and tested communication protocols between AV fleet managers and emergency dispatch would reduce the coupling failures that have caused emergency response delays.
2. Testing vehicles on degraded infrastructure scenarios such as dark signals and no cellular connection, as well as removing remote operator confirmation within defined safety thresholds.

Human System Redesign

1. Establish a maximum vehicle-to-operator ratio (e.g., 25:1) with mandatory surge capacity during declared local emergencies to minimize cognitive load.

Governance Restructuring

1. Transfer or share commercial deployment permitting authority with affected municipalities above a fleet threshold. A city with 1,000 autonomous vehicles on its roads should have formal regulatory capabilities.
2. Require robotaxi operators above a fleet size threshold to conduct annual mock disaster drills with city officials, demonstrating fleet recall, emergency routing, and operator surge capacity before the next California earthquake.

Summary

The deployment of robotaxis in San Francisco robotaxi is a study of a technically sophisticated system deployed without adequate design of the social, organisational, and governance systems affected by it. Waymo and Cruise operated commercial AV fleets in one of the world's most complex urban environments, navigating a fragmented regulatory structure split across the California DMV, the CPUC, and the City of San Francisco with no direct regulatory authority. Systems psychology analysis identifies a joint optimization deficit as the root cause, highlighting the development and deployment of the technical subsystem with minimal co-design of the social systems. This produced failures at various system boundaries; between vehicles and pedestrians who had no social signal to read, between fleet operators and regulators who received selectively managed safety information, and between AV operations and emergency services who had no shared protocol. Causal loop analysis shows three reinforcing dynamics of technology learning, fleet scaling risks, and trust destruction, and two balancing loops of regulatory correction and public acceptance.

Conclusion

San Francisco's robotaxi system is one of the most significant deployments of artificial intelligence in commercial operation. Robotaxi vehicles navigate a complex urban environment, serve hundreds of thousands of passengers per week, and accumulate safety data at a tremendous scale. However, systems analysis reveals that the vehicle is only one node in a larger, more fragile network of remote operators who cannot scale as fast as the fleet, urban infrastructure that was not designed with fleet-scale autonomous vehicles in mind, a regulatory architecture that was built for a different era of transportation, and a city that bears the operational consequences of decisions made at the state level. The December 2025 blackout was not primarily a technology failure, but a systems failure. It was a product of a reinforcing loop of rapid fleet growth running faster than the balancing loops of regulatory adaptation, remote operator capacity, and infrastructure resilience. The path to a mature, safe, and legitimate robotaxi system in San Francisco must include redesigning the human, regulatory, and infrastructure systems that surround it. The vehicle must become less dependent on external systems, achieving system failure consequences that remain bounded, local, and recoverable.

References

Fraedrich, E., Beiker, S., & Lenz, B. (2015). Transition pathways to fully automated driving and its implications for the sociotechnical system of automobility. European Journal of Futures Research, 3(1), 1–11. https://doi.org/10.1007/s40309-015-0067-8

Grigorescu, S., Trasnea, B., Cocias, T., & Macesanu, G. (2020). A survey of deep learning techniques for autonomous driving. Journal of Field Robotics, 37(3), 362–386. https://doi.org/10.1002/rob.21918

Jomehpour Chahar Aman, J., & Smith-Colin, J. (2020). Transit Deserts: Equity analysis of public transit accessibility. Journal of Transport Geography, 89, Article 102869. https://doi.org/10.1016/j.jtrangeo.2020.102869

Meir, A., Grimberg, E., & Musicant, O. (2025). The human-factors' challenges of (tele)drivers of Autonomous Vehicles. Ergonomics, 68(6), 947–967. https://doi.org/10.1080/00140139.2024.2346552

Şahin, H., Hemesath, S., & Boll, S. (2022). Deviant Behavior of Pedestrians: A Risk Gamble or Just Against Automated Vehicles? How About Social Control? Frontiers in Robotics and AI, 9, 885319. https://doi.org/10.3389/frobt.2022.885319

Schrank, A., Wilbrink, M., Brandenburg, S., & Oehl, M. (2025). Improving road user perception in adverse weather: An augmented human–machine interface for remote assistants of automated vehicles. Transportation Research. Part F, Traffic Psychology and Behaviour, 113, 500–516. https://doi.org/10.1016/j.trf.2025.05.017

Svenson, O., & Eriksson, G. (2017). Mental models of driving and speed: biases, choices and reality. Transport Reviews, 37(5), 653–666. https://doi.org/10.1080/01441647.2017.1289278

Synopsys Editorial Staff. (2025, February 15). The 6 levels of vehicle autonomy explained. Synopsys. https://www.synopsys.com/blogs/chip-design/autonomous-driving-levels.html

Wikipedia. (2025, February 19). Waymo. In Wikipedia, The Free Encyclopedia. https://en.wikipedia.org/w/index.php?title=Waymo&oldid=1341651683